Trigger Points and Myofascial Pain: From Physiology to Pain Experience

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Fall 2025

Objectives:

- Differentiate between nociception (the physiological signal) and pain (the subjective experience) in the context of myofascial pain.
- Describe the pathophysiology of a trigger point, including the role of the neuromuscular junction, the energy crisis model, and how a local lesion can drive a broader pain experience.
- Explain the neurological connection between trigger points and the central nervous system, including the concept of central sensitization.
- Understand the mechanism behind treatments, including osteopathy

Pain vs. Nociception

- **Nociception:** physiological process of detecting a noxious stimulus (nerve-level signal).
- **Pain:** subjective, multidimensional experience shaped by sensory, emotional, and cognitive factors.
- Trigger points are **sources of nociceptive input**, but patients feel **pain as a broader lived experience**.
- Important distinction: treatment must address both the *signal* and the *experience*.

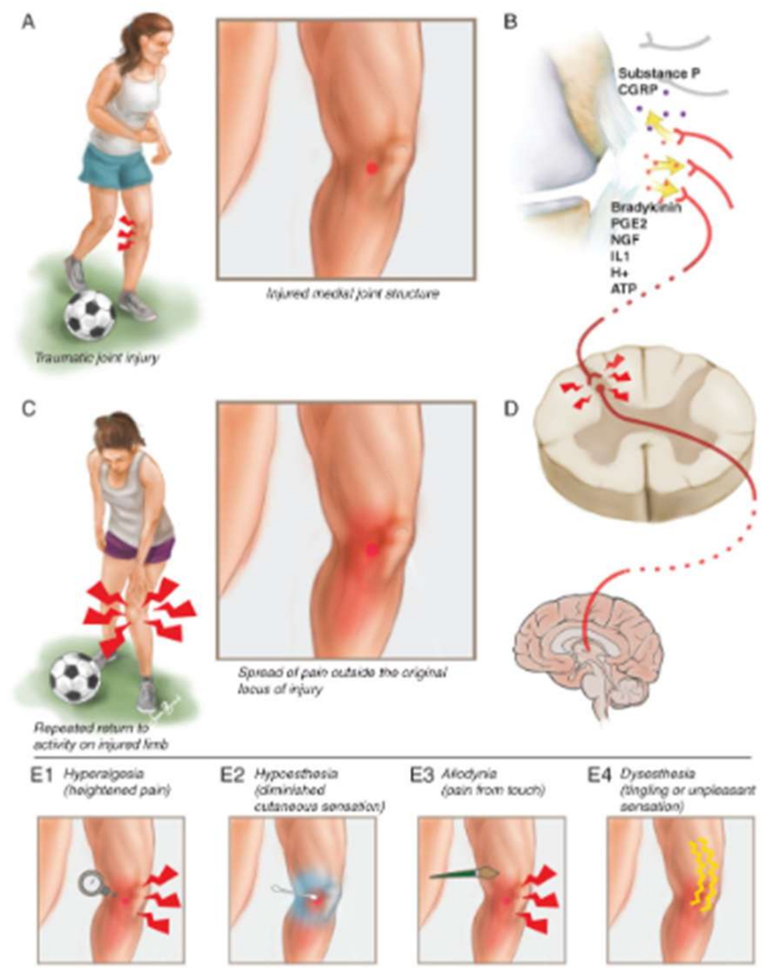


Figure 1-13. Peripheral and central sensitization. A, Injury to the knee joint. B, Nociceptive response to the injury. C, Peripheral sensitization. D, Secondary hyperalgesia. E1-E4, Quantitative sensory testing to differentiate peripheral and central sensitization.

Trigger Points Defined

Clinical definition: *Hyperirritable spot within a taut band of skeletal muscle.*

Painful on compression; may reproduce *referred pain patterns*.

Etiologic definition: *Dysfunctional motor endplates with localized contracture knots.*

Serve as a **persistent driver of nociceptive input** → influences local and distant pain

Can progress to **Myofascial Pain Syndrome (MPS)**

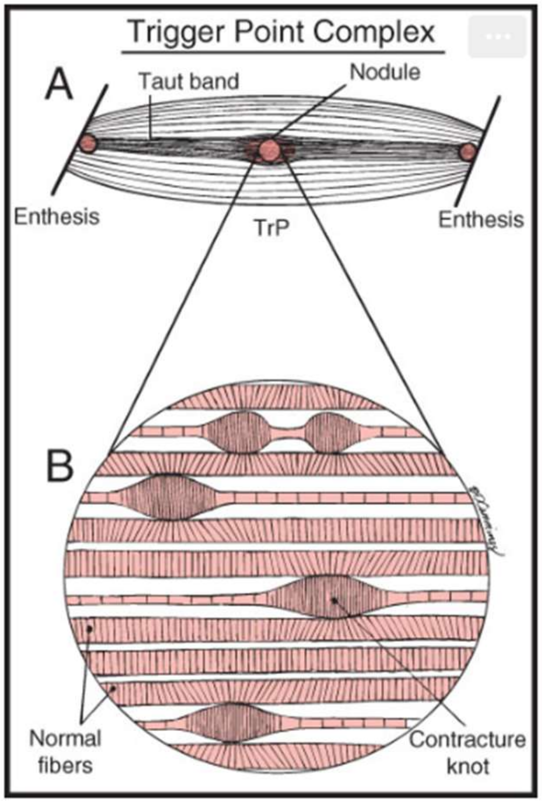
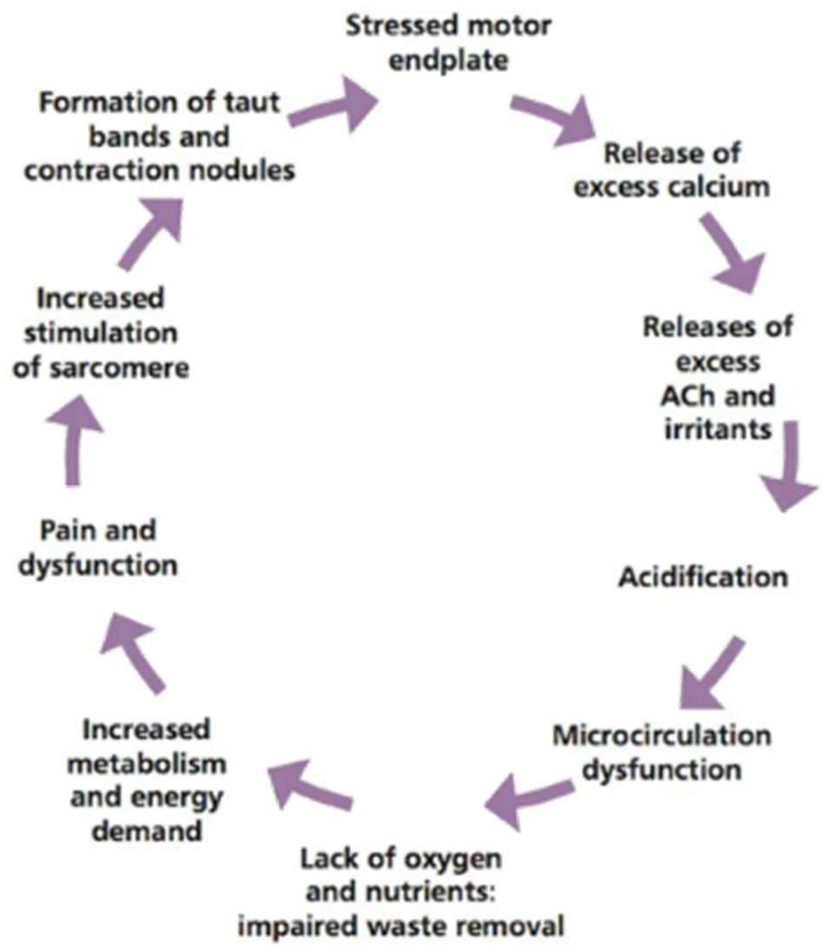


Figure 2-8. Schematic of a TrP complex of a muscle in longitudinal section. A, The TrP which is in the endplate zone and contains numerous contracture knots. B, Enlarged view of part of the TrP shows the distribution of five contracture knots and their effects on adjacent sarcomeres.

The Vicious Cycle (Overview)



Cycle of misery: trigger point formation hypothesis – these individual links in the chain do not always occur in this order. (Sourced from: Starlanyl & Sharkey 2013.)

The Initial Insult

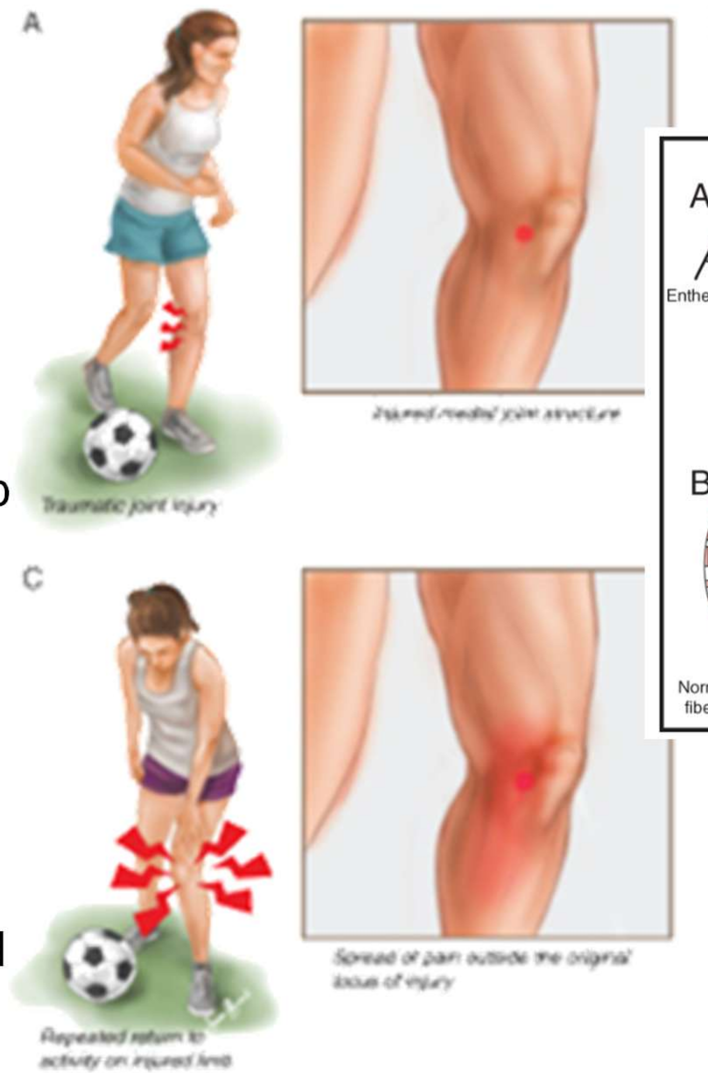
Triggered by acute trauma (injury) or chronic overload (posture, repetitive strain)

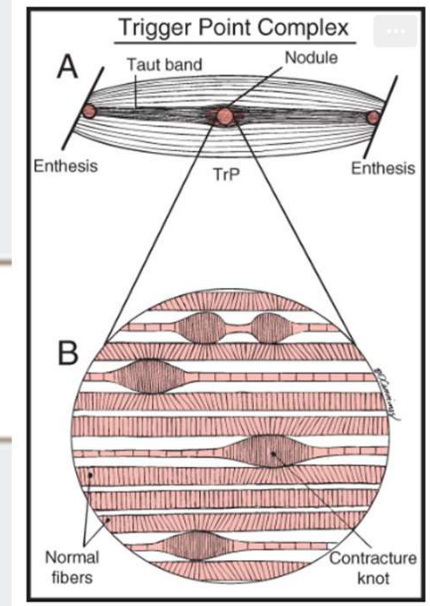
Muscle overload exceeds local metab supply

Sustained contraction → ischemia, impaired perfusion, energy crisis

Localized dysfunction leads to a taut band / trigger point formation

Fascia plays a role in distributing load and transmitting mechanical stress





Normal Neuromuscular Junction

Motor neuron action potential reaches axon terminal

Calcium influx triggers release of **acetylcholine (ACh)**

ACh binds receptors on muscle fiber → depolarization (endplate potential)

Muscle action potential spreads along sarcolemma → contraction

ACh broken down by acetylcholinesterase → allows relaxation and reset

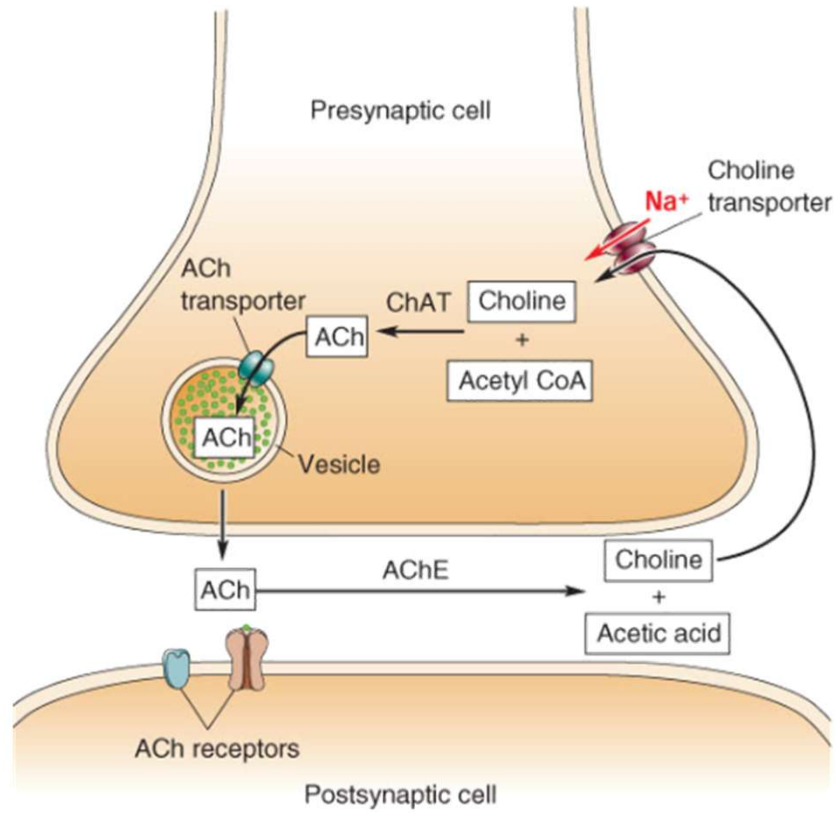
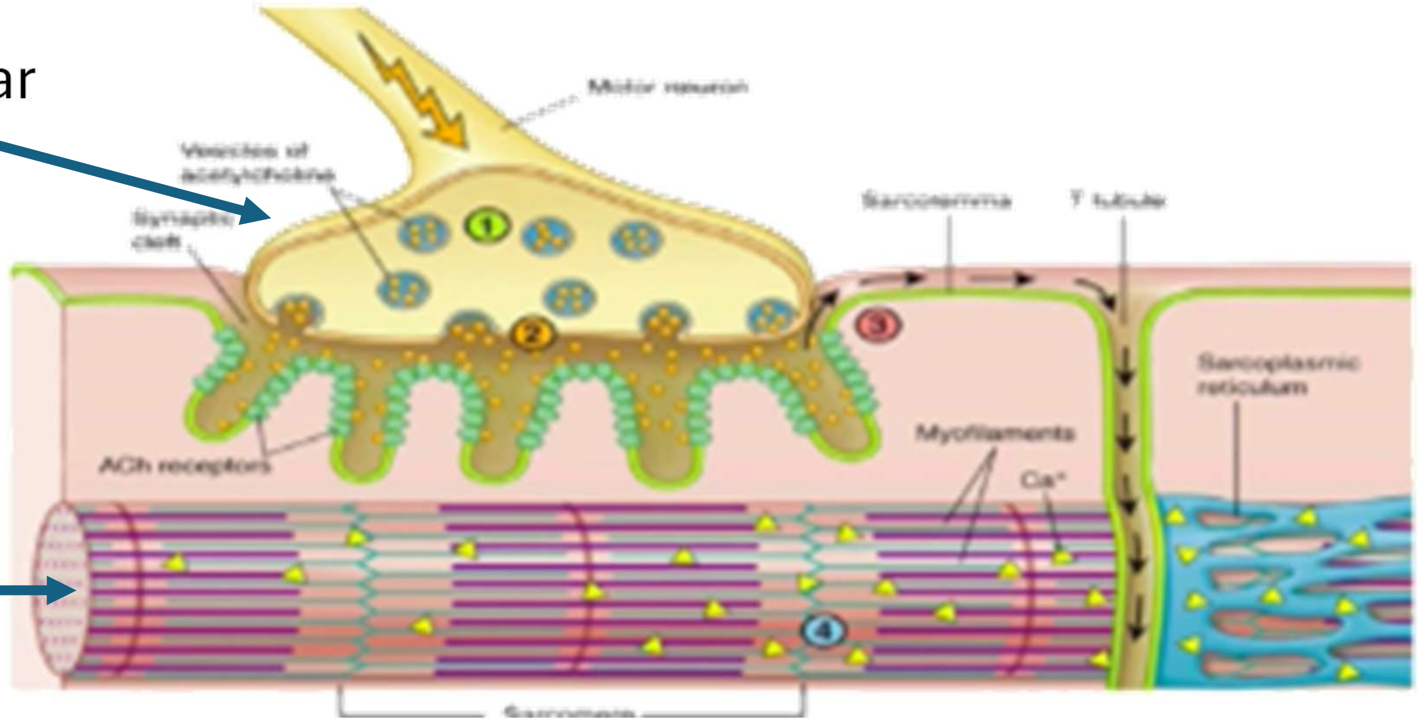


Figure 1-18. The life cycle of acetylcholine (ACh).

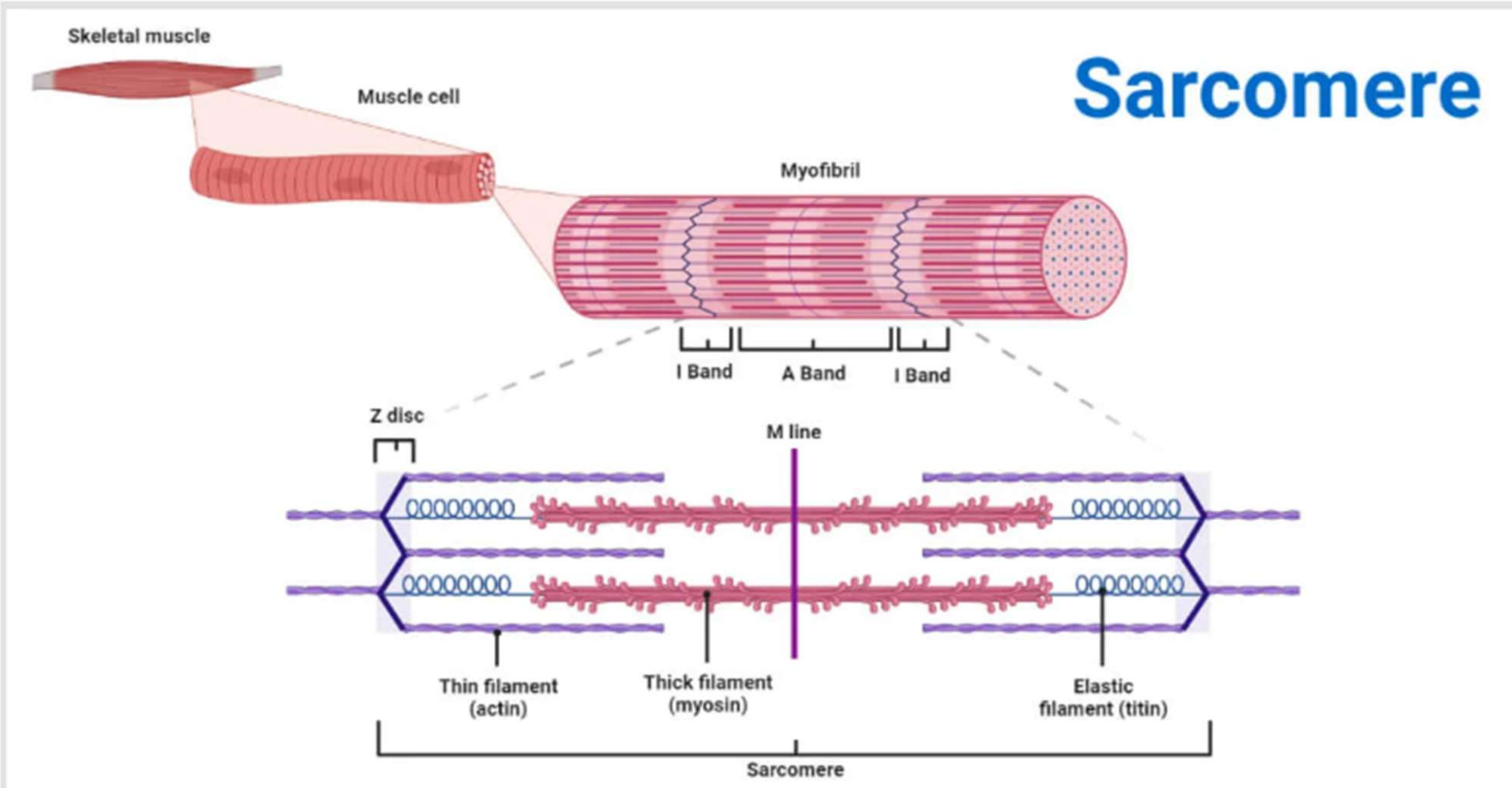
Depolarization → Actin and myosin contraction in the sarcomere

Neuromuscular junction

Sarcomere



sarcomere stays contracted


Skeletal muscle
Sarcomere
Muscle cell
Myofibril
I Band
A Band
I Band
Z disc
M line
Thin filament
(actin)
Thick filament
(myosin)
Elastic
filament (titin)
Sarcomere

Neuromuscular Junction Dysfunction

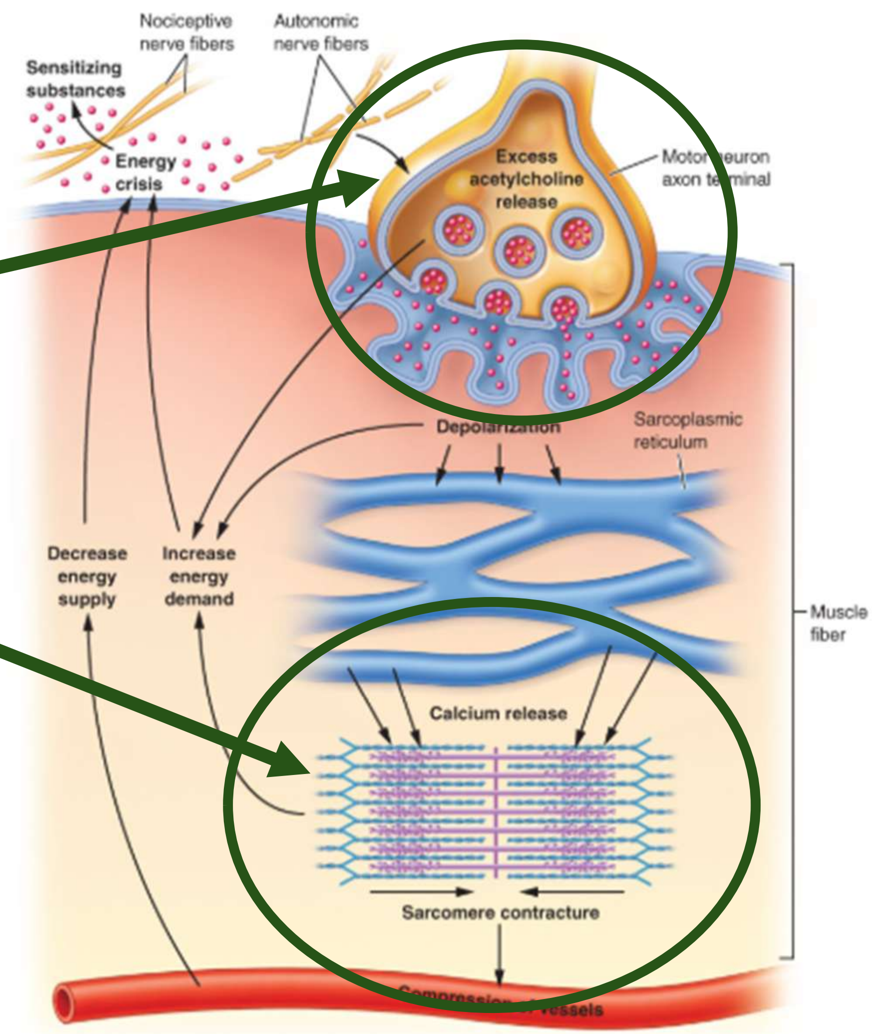
Excess acetylcholine (ACh) release at the NMJ

Persistent depolarization of muscle fiber

Leads to **sustained sarcomere contraction** → contracture knot

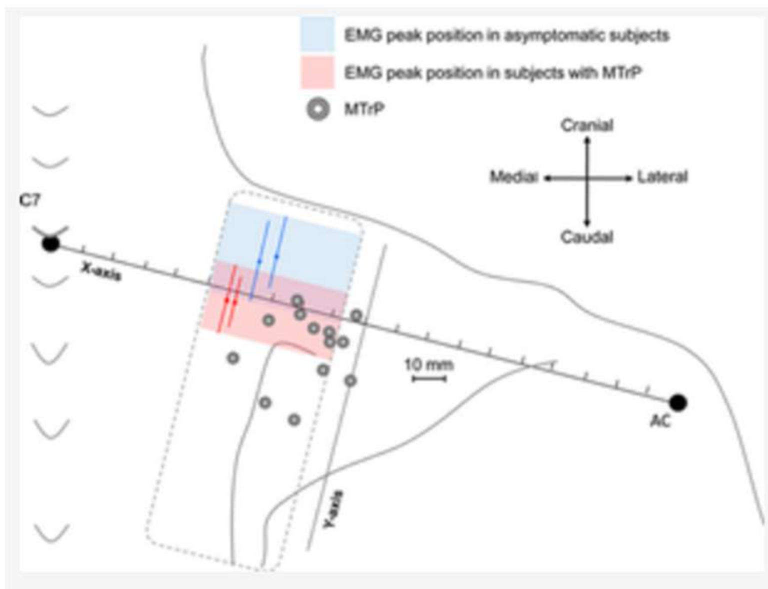
Appears on EMG as **endplate noise** (physiologic marker of trigger points)

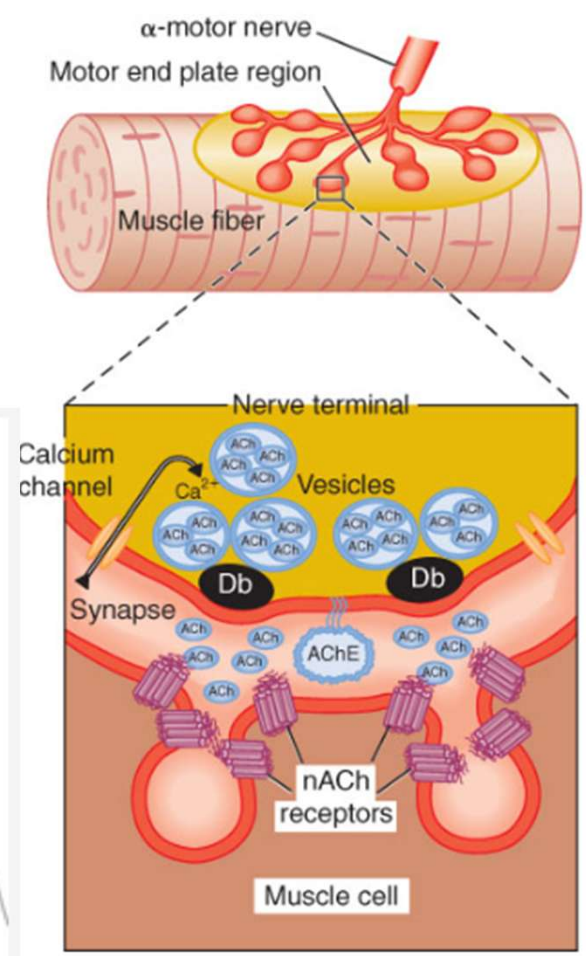
Figure 2-10. Integrated hypothesis. The primary dysfunction hypothesized here is an abnormal increase (by several orders of magnitude) in the production and release of acetylcholine packets from the motor nerve terminal under resting conditions. The greatly increased number of miniature endplate potentials produces endplate noise and sustained depolarization of the postjunctional membrane of the muscle fiber. This sustained depolarization could cause a continuous release and uptake of calcium ions from local sarcoplasmic reticulum and produce sustained shortening (contracture) of sarcomeres. Each of these four highlighted changes would increase energy demand. The sustained muscle fiber shortening compresses local blood vessels, thereby reducing the nutrient and oxygen supplies that normally meet the energy demands of this region. The increased energy demand in the face of an impaired energy supply would produce a local energy crisis, which leads to the release of sensitizing substances that could interact with autonomic and sensory (some nociceptive) nerves traversing that region. Subsequent release of neuroactive substances could in turn contribute to excessive acetylcholine release from the nerve terminal, completing what then becomes a self-sustaining vicious cycle.



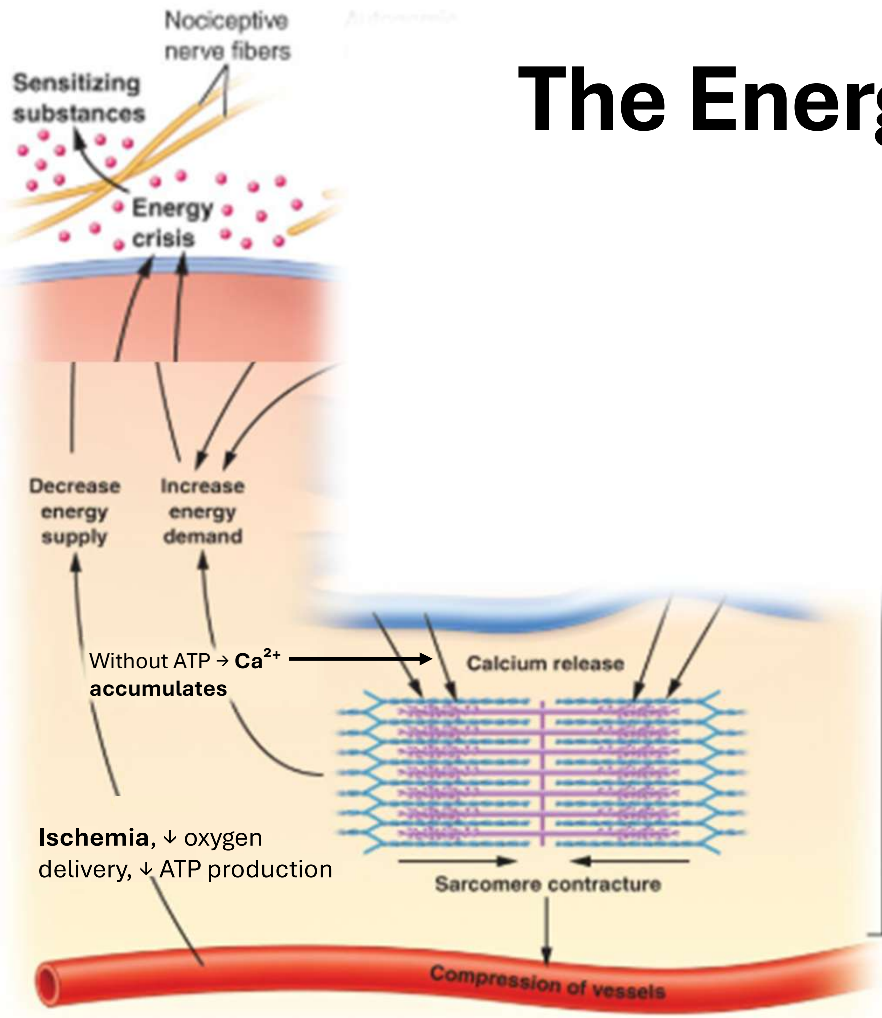
End plate noise

Appears on EMG as **endplate noise** (physiologic marker of trigger points)





The Energy Crisis Model



Sarcomere contracture →

Compression of vessels →

↓ acetylcholinesterase AND

Ischemia, ↓ oxygen delivery, ↓ ATP production →

ATP required for Ca^{2+} reuptake into sarcoplasmic reticulum →

Without ATP → Ca^{2+} accumulates, sarcomeres remain contracted →

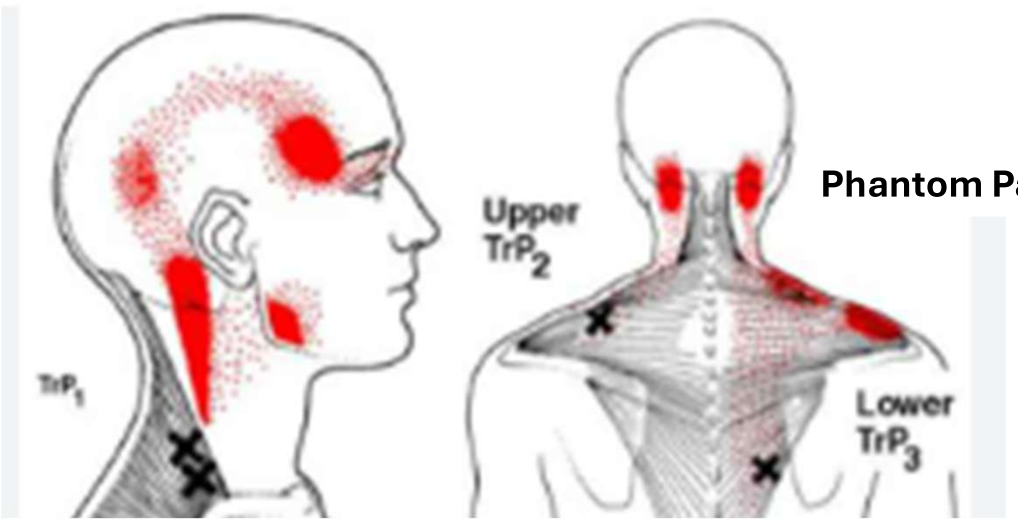
↓ Energy Supply + ↑ Energy demand

= **ENERGY CRISIS MODEL**

why trpts persist

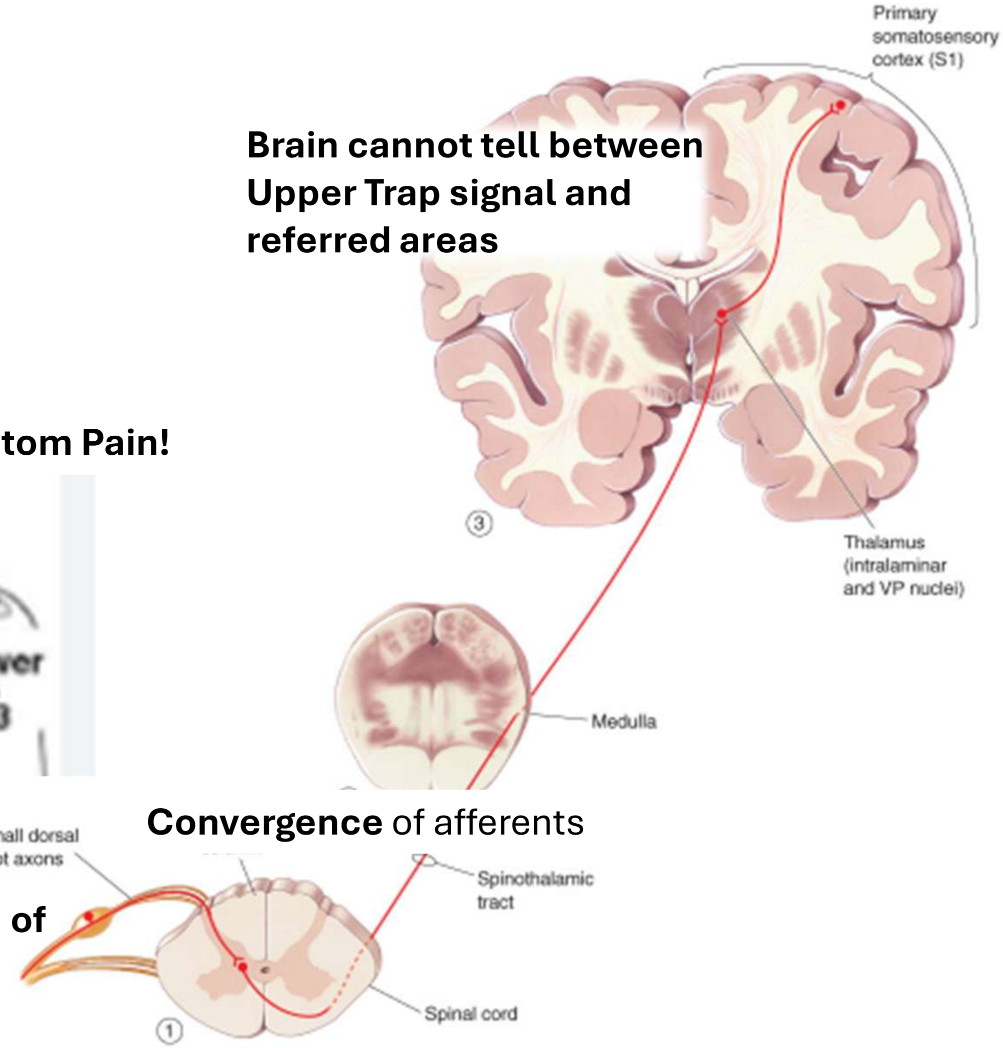
Referred Pain:

Brain cannot tell between Upper Trap signal and referred areas



Phantom Pain!

XX = TrP nociceptive input



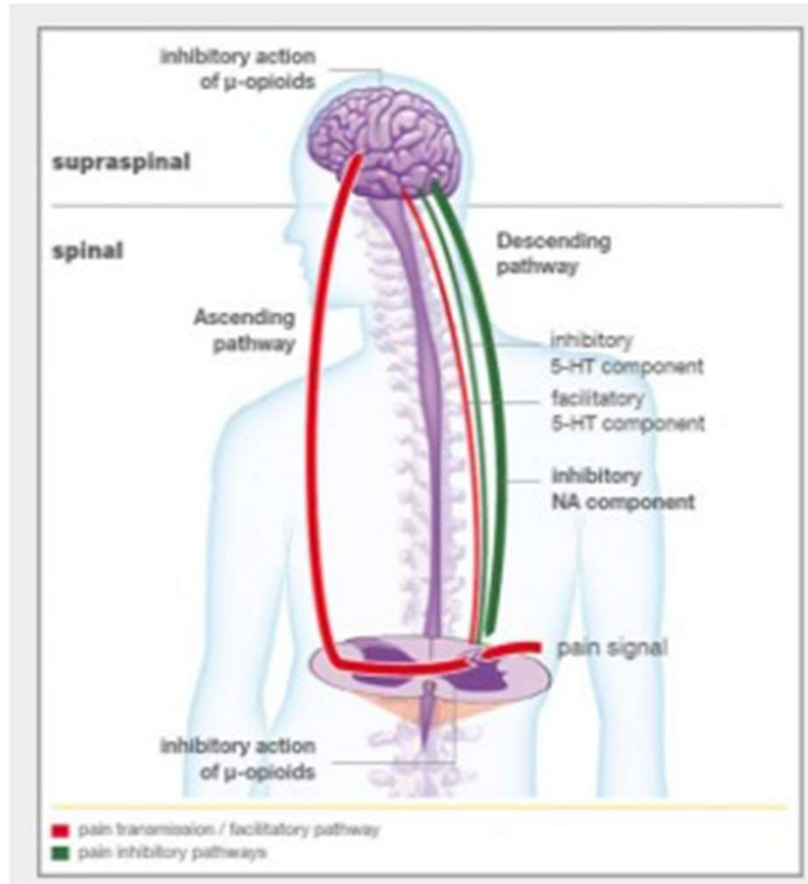
Convergence of afferents

Message sent to dorsal horn of spinal cord

Figure 1-5. The spinothalamic pathway. This is the major route by which pain and temperature information ascend to the cerebral cortex. (From Bear MF, Connors BW, Paradiso MA. *Neuroscience: Exploring the Brain*. 4th ed. Philadelphia, PA: Wolters Kluwer; 2016.)

Central Sensitization

- Persistent nociceptive input → sensitized dorsal horn & brain pathways
- **Inhibitory interneurons fail** → loss of normal "braking" of pain signals
- Neurons fire more easily → amplified signaling
- Clinical signs:
  - **Hyperalgesia** = exaggerated pain to painful stimulus
  - **Allodynia** = pain from non-painful input
- **"Software not hardware problem":** Alarm keeps ringing even when fire is gone



pain is usual a signal of tissue dama

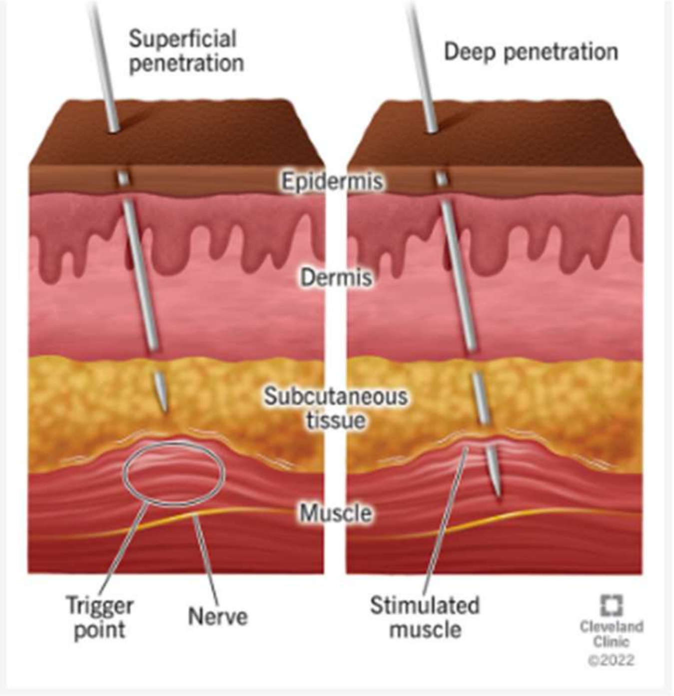
Dry needling/injection -> disrupt knot, elicits twitch, resets sarcomere, decreases Ach release
compression/stretch -> brief ischemia + reperfusion restores O2/ATP; stretching normalizes endplate activity, decreases noise
OMT & Manual therapies -> decrease nociceptive input and dorsal horn sensitization; decrease sympathetic drive; increase endorphin release; increase circulation

stuck in the past and tx bringing them to the present

Mechanism of Treatment

Dry needling / Injection

- Mechanical disruption of contraction knot
- Local twitch response
- Rapid release and reuptake of Ca^{2+}
- Resets sarcomere length
- Normalizes ACh release



Mechanism of Treatment

- **Ischemic compression / Inhibition** → Temporarily reduces blood flow; reperfusion restores O_2 & ATP → Ca^{2+} reuptake → relaxation
- **Stretching / Spray-and-stretch** → Normalizes endplate activity; elongates taut band; decreases endplate noise

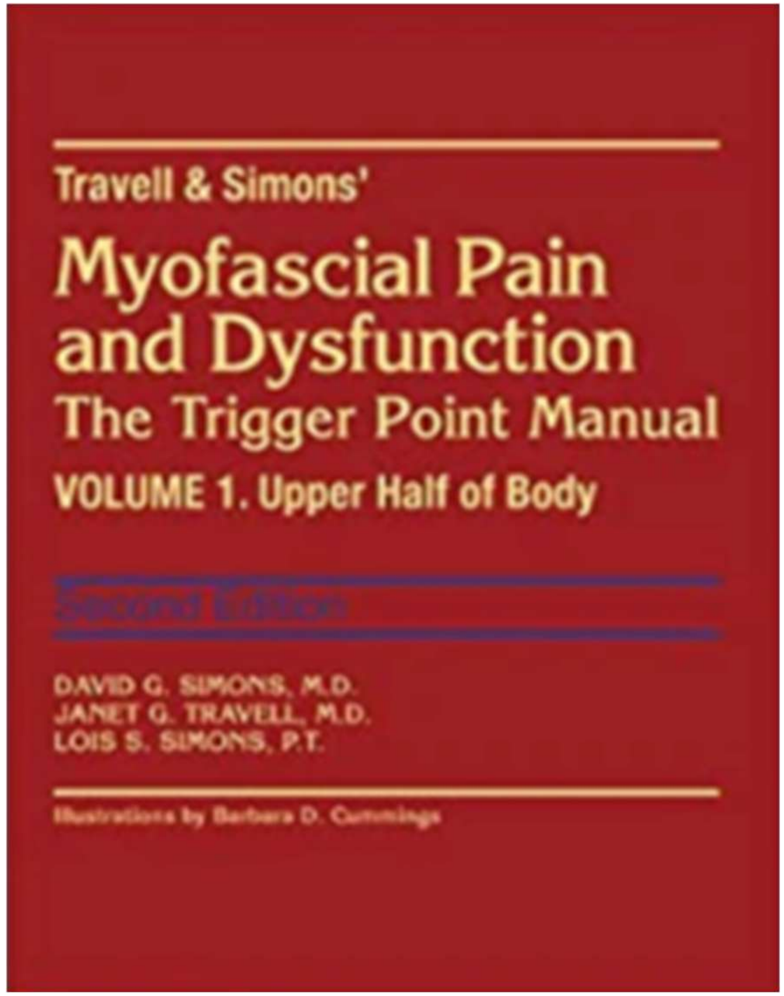
Mechanism of Treatment

- **OMT & manual therapies** →
- ↓ nociceptive input
- ↓ dorsal horn sensitization
- ↓ sympathetic drive
- ↑ endorphin release (decreases nociceptive input and sensitization)
- ↑ circulation and ↓ ischemia
- "Reboots" the whole system to the present moment (new pattern)

Clinical Takeaway

- **Nociception vs Pain**
  - – Nociception = nerve signal
  - – Pain = lived, multidimensional experience
- **Trigger Point Physiology**
  - – NMJ dysfunction + energy crisis → contracture knot
  - – Local lesion can drive widespread pain
- **Nervous System Connection**
  - – Persistent nociception → central sensitization
  - – Explains hyperalgesia & allodynia
- **Treatment Mechanisms**
  - – OMT and manual therapies reduce nociception, restore circulation, rebalance nervous system

References


Travell & Simons'
Myofascial Pain and Dysfunction
The Trigger Point Manual
VOLUME 1. Upper Half of Body
Second Edition
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